

UR5e + DexHand v2 Apple-Picking Gripper

Design summary, debugging timeline, best results, and novelty analysis

Author: Mahimaa Priyadharshini S. Period: 23 Jul – 14 Sep 2026 Report date: 14 Sep 2026 Covers commits: initial → c7399dc

1. Objective

Pick an apple off a table with a simulated UR5e arm and DexHand v2 hand, in Gazebo, without bruising, dropping, or flinging it — using only the sensing a real robot would have (joint states, TF, a camera), never the simulator's ground truth. A local vision-language model looks at the apple first and estimates how gently it should be handled.

2. Abstract

A UR5e on a mobile base, fitted with a 21-joint DexHand v2, picks apples from a table in Gazebo/ROS 2. A gripper-mounted camera feeds a local Qwen2.5-VL model (via Ollama), which estimates fragility and recommended grip force; that estimate drives a friction-cone force calculation that sets closing force and finger travel. Almost none of the real difficulty was in the vision layer — it was in getting the arm to reach the right pose and the hand's fingers to actually meet the apple's surface. Most of this report is that debugging process.

Status as of 14 September. All ten apples have been picked up and held using a single set of grasp settings: 29 of 30 attempts. The one miss was a frozen simulation, not a failed grasp, and no apple slipped in any attempt. The apples span 10.0–12.0 cm in diameter, 0.40–0.70 kg in mass and 0.70–0.95 in friction. Every pick lifted the apple +14.2 to +15.3 cm on a 15.0 cm lift, and neither per-apple scaling nor a learned policy was needed. The first complete grasp came on 7 September. Reliability came on 13–14 September from finding that several apparent grasp and arm failures were timing artefacts. Gazebo runs at 1.5–9% of real time on this machine and sometimes freezes, so the arm was being measured and corrected while still moving, and lifts were judged before they had finished. Once every wait used the simulation's own clock, the arm landed on its target with no corrections, and the pick rate on the tuning apple went from 3 of 4 to 8 of 8. The working settings have not yet been carried into the main vision-driven pipeline.

3. Tech Stack

Layer	Used	Why
Simulator	Gazebo (Ignition/Fortress)	Real physics so a bad grasp fails like a real one would.
Middleware	ROS 2 Humble, ros2_control, TF2	Joint control, state, and coordinate transforms.
Arm	UR5e (6-DOF) + custom IK on ikpy	Needed heavy tuning to stop it picking wrong solutions.
Hand	DexHand v2, 21 joints / 5 fingers	Enables a pocket/power grasp, not just a pinch.
Planning	MoveIt 2	Available alongside the custom IK path.
Vision-language	Qwen2.5-VL (3B) via Ollama, local HTTP	Fragility / material / grip-force estimate per apple.
Code	Python 3, NumPy, ikpy, OpenCV, cv_bridge	All grasp logic and diagnostics — no custom C++ nodes.
Dev environment	WSL2 (Ubuntu) on Windows, VS Code, Git	Runs the Linux-only ROS 2/Gazebo stack from Windows.

4. Prior Work Used

- UniversalRobots/Universal_Robots_ROS2_Description — UR5e URDF/kinematics.
- UniversalRobots/Universal_Robots_ROS2_GZ_Simulation — base Gazebo launch/config.
- iotdesignshop/dexhandv2_description — DexHand v2 URDF; source of most hand-geometry bugs (Section 6).
- touchlab-avtarx/allegro_hand_description — kept as an alternate hand, unused in the final pipeline.
- PickNikRobotics/ros2_robotiq_gripper — alternate end-effector reference.
- ros-controls/gz_ros2_control, ros2_control, ros2_controllers — the joint control stack.
- gazebo-sim/ros_gz — ROS 2 ↔ Gazebo bridge. moveit/moveit2 — motion planning.
- wjwwood/serial (via tylerjw/serial) — serial dependency of the gripper driver.

Everything in `src/my_pick_and_place/scripts` (~25 files: IK correction, grasp pipeline, geometry/station measurement tools, closing-policy search, VLM node) and `tools/` is original to this project. None of the repos above ship any grasping logic of their own.

5. How the System Works

5.1 Basic process, one attempt end to end

- Reset.** Arm returns to a fixed rest pose; the target apple is teleported back to its known spawn position. The attempt only begins once four consecutive position readings show it still and on its spot; otherwise it is re-teleported, and respawned if it keeps its velocity.

- 2. **Locate.** The apple's live pose is read from its Gazebo pose topic and sanity-checked against the table bounds, so a corrupted/flung position aborts the attempt instead of wasting it.
- 3. **Aim.** IK is solved twice: once roughly, to learn which way the wrist will face, then again with the target corrected so the measured fingertip pocket — not the bare wrist — lands on the apple. The palm is tilted 45° from flat, and the apple is centred across the four fingers rather than all five.
- 4. **Approach.** The arm moves toward a pose 0.16 m back along the hand's forward axis, then to the grasp pose. The code waits until the grasp move has finished in simulated time, then reads the real wrist position from TF. If the wrist is more than 12 mm off, up to three damped corrections follow; since 14 September none have been needed, as the arm lands within 1 mm. The apple's position is checked after every arm move, and any disturbance is logged against the move that caused it.
- 5. **Grasp.** The 7-layer sequence below runs (perceive → plan → close → lift → check safety).
- 6. **Place.** If the lift and safety checks pass, the arm carries the apple to a delivery station and releases it into a crate; if it settles more than 0.3 m from the crate centre, the attempt is marked *missed* even though the grasp itself worked.
- 7. **Log.** Every attempt — success or failure — is appended to `experience_log.json`, and the test run ends with a results table covering every attempt.

5.2 The 7-layer grasp architecture

#	Name	What it does
1	Perception	Captures a gripper-camera frame and sends it to the local Qwen2.5-VL model; gets back a fragility score, material guess, and recommended grip force.
2	Imagination	Converts the fragility score into a minimum grip force (friction-cone calculation: $\text{mass} \times g \div 2\mu$, scaled by a safety margin) and a maximum finger-pitch ceiling — the plan, computed before any motion happens.
3	Motor execution	The thumb closes first, to first contact and then a force-capped preload, so the fingers have something to close against; the four fingers then curl inward in small steps. Revised 7 Sep: thumb-last was measured to let the side-by-side fingers push the apple out before the thumb arrived.
4	Force feedback	After every step, reads each finger's joint effort; a finger is marked "contacted" once its effort crosses a threshold calibrated per finger at the start of every run, cross-checked against the fingertip's real TF position relative to the apple's live position.
5	Smooth lift	Squeezes past first contact, then raises the wrist 0.15 m. A result counts as held only if the apple rises at least half that and ends within 0.20 m of the wrist. The lift is judged only after the arm has had its whole commanded move in simulated time; if the simulation clock stops, the attempt is reported as a frozen simulation rather than as a grasp result.
6	Safety monitor	Watches finger effort after the lift; if any finger crosses a danger threshold, opens the hand and aborts the hold instead of risking a crush.
7	Experience logging	Records the full attempt — VLM read, grip plan, contact results, before/after apple pose, reward — to <code>experience_log.json</code> , whether it succeeded or not.

Between Layer 6 and Layer 7, an unnumbered *Place* step (5.1, step 6) carries the apple to the crate if the grasp and safety checks passed. It isn't counted as one of the "7" layers in the code, but it's the difference between a held apple and a delivered one.

5.3 Tests and diagnostic tools written for this project

Beyond the main pipeline, this project accumulated a set of small, purpose-built scripts — each written to answer one specific question rather than to run the full task.

Reachability & station diagnostics

Script	What it tests
<code>find_station.py</code>	Sweeps candidate mobile-base positions and reports the real IK-solved shoulder-pan mismatch for each.
<code>arm_table_clearance_check.py</code>	Checks real per-link clearance above the table, to tell a genuine collision apart from the arm straining against gravity.
<code>shoulder_lift_sweep.py</code>	Isolates whether shoulder-lift's 150 Nm saturation is caused by contact or by gravity/lever-arm alone.
<code>grasp_height_sweep.py</code>	Tests several wrist heights against real measured finger geometry to find one where the fingers reach the apple's body, not just its top.

Hand-geometry diagnostics

Script	What it tests
<code>finger_geometry_check.py</code>	Measures real fingertip position relative to the wrist via TF, open and fully closed — the tool that found the 9.11 cm closed span.
<code>finger_joint_probe.py</code>	Checks which finger joints actually appear in <code>/joint_states</code> and actually move when commanded — caught the dead-mimic-tag bug.
<code>thumb_opposition_check.py</code>	Sweeps thumb yaw/roll and measures where the thumb tip ends up relative to the palm. Its numbers were the key to the 7 Sep thumb diagnosis.

Grasp-quality tests

Script	What it tests
--------	---------------

<code>diagnostic_grasp_attempt.py</code>	One attempt with the most forgiving settings reasonably justifiable, to tell whether a 3+ finger grasp is structurally possible at all.
<code>pocket_grasp_test.py</code>	The main experiment. Per attempt it now measures each fingertip's gap to the apple, the height at which each finger meets it relative to its equator, the thumb's angle to the fingers about the apple, clearance to the camera body, and the apple's movement after every arm move; it ends every run with two results tables and a per-setting summary. It also watches the whole lift, sampling hand height, apple height, per-finger load, thumb load and all six arm joint efforts against their limits, timed in simulated rather than wall-clock time; supports empty-hand control attempts (the same attempt with the apple removed); and scores an attempt where the apple stayed in the hand but the hand did not rise separately from one where the apple was dropped.
<code>test_all_apples.sh</code>	Runs <code>pocket_grasp_test.py</code> on a list of apples in sequence with a chosen number of attempts, keeps each full log, and prints one summary line per apple. Used for the all-apples result (Table M).
<code>train_closing_policy.py</code>	Small (1+λ) evolution-strategy search over 10 per-finger closing parameters, scored by a reward tied to real contact, effort, and lift height.
<code>pick_all_apples.py</code>	Multi-station orchestration across all 10 apples.

Perception, visualization and code quality

Script	What it does
<code>detect_apples.py</code>	Colour-thresholding apple detector on the overhead camera, with a <code>--calibrate</code> mode printing detection against live ground truth.
<code>rviz_world_markers.py</code>	Publishes apples and table as RViz markers using the same pose topics and coordinate math the grasp code uses.
<code>tools/check_scopes.py</code>	Finds names used but never defined, per function scope, using Python's <code>symtable</code> . Written after the same class of <code>NameError</code> reached live runs three times; run over every script before each commit. Extended on 14 Sep to flag a local variable that hides a module-level function of the same name, after that caused an <code>UnboundLocalError</code> in a live run.

Earlier pick scripts (superseded)

`direct_pick.py`, `pick_block.py`, `smart_pose_picker.py`, `auto_block_tracker.py`, `adaptive_picker.py`, `ik_sweep_picker.py`, `sweep_search_picker.py`, `moveit_pick_place.py`, `pick_and_place.py` — early single-object pickers built against a placeholder block, several relying on a simulated `/attach` "glue" topic rather than real friction. Superseded once `full_layer_grasp.py`'s real, feedback-driven grasp existed, but kept as working reference points. `object_geometry.py` is a shared helper that reads an object's true dimensions from its SDF rather than hardcoding them.

6. Timeline — Bugs Found and Fixed

One continuous record from the first working gripper to the latest commit. Every entry is a real defect found in a live run, not a planned feature. Hypotheses that were tested and found wrong are recorded as such.

23–27 Jul — first working gripper

Date	Bug	Fix
26 Jul	Hand loaded from a URDF variant with no real revolute joints — fingers never moved.	Switched to <code>dexhandv2_right.urdf</code> , the variant with working joints; merged its controller config.
26 Jul	Arm inverse-kinematics math was wrong.	Corrected the IK calculation.
26 Jul	"Attach" (finger-closing) function had logic errors.	Fixed and improved the attach mechanism.

29 Jul – 4 Aug — a test bed worth trusting

Date	Bug	Fix
31 Jul	World file's camera sensor plugin wasn't Ignition-compatible; launch arg was wrong.	Fixed launch arg, swapped in an Ignition-compatible camera sensor.
31 Jul	World depended on an external Fuel-hosted ground plane/sun model — broke if offline.	Inlined sun and ground plane directly in the world file.
3 Aug	Apple world had duplicate ground planes and stale Fuel model URIs.	Removed the duplicate, fixed the URIs.
3 Aug	Mobile base spawned in a pose that didn't suit apple picking.	Updated base pose and initial arm posture.

27 Aug — reach was often wrong, and Gazebo itself was crashing

Date	Bug	Fix
27 Aug	Gazebo crashed on certain graphics settings (Ogre2/GL3Plus).	Switched sensor rendering to legacy Ogre1; ran headless by default.
27 Aug	Hardcoded paths and a joint-limit bug broke multi-station runs.	Fixed paths/limits; added multi-station pick-and-place.
27 Aug	IK solver too narrow — many real targets reported unreachable.	Widened error tolerance, added far more starting guesses.
27 Aug	Grasp target aimed at the apple's centre instead of a sensible hand height.	Corrected the target height calculation.
27 Aug	Mirror arm configuration: solver kept picking the flipped, backwards twin of the correct pose.	Always select the solution whose shoulder-pan angle is closest to the geometrically expected one.
27 Aug	Five DexHand links had zero inertia, producing NaN joint-effort readings.	Fixed the inertia values; added the missing effort-sensing interface to all 21 hand joints.

28 Aug — the mirror-configuration bug, properly diagnosed

Date	Bug	Fix
28 Aug	Root cause 1: the UR5e URDF inserts a mandatory 180° Z-rotation joint between base and arm; the IK chain-builder skipped straight past it, so every shoulder angle was solved in a frame rotated 180° from reality — a consistent ~165–172° pan offset.	Added the missing <code>base_link-base_link_inertia</code> joint into the IK chain's <code>base_elements</code> .
28 Aug	Root cause 2: DexHand's URDF branches at the wrist into 5 fingers + camera; ikpy silently treated the first branch (thumb) as the end effector, solving IK to place the <i>thumb tip</i> at the target — a ~13 cm error on a rotated axis.	Programmatically pruned every joint/link downstream of the hand's root link before building the chain, so the wrist is the unambiguous last link.
28 Aug	A hard shoulder-pan lock and a full-orientation constraint (earlier workarounds) were making genuinely reachable targets unreachable.	Removed both now that the real causes were fixed; pan is a soft-preferred seed, not a hard constraint.
28 Aug	Commanded shoulder_lift of ~193° physically stuck at ~213° pulling 71 Nm — a "valid" IK solution swinging the arm through its own body.	Added a tighter swing-angle cap ($\pm 135^\circ$) specifically on shoulder_lift.
28 Aug	Camera rig and self-collision settings were physically inside the arm's own working volume.	Moved the camera out of the way; disabled unhelpful self-collision between arm and base.
28 Aug	A stale hardcoded offset between the robot's base and arm mount was skewing every position.	Removed the offset; re-tuned the pick station to (1.25, -0.90).
28–29 Aug	Code trusted commanded joint angles instead of checking what the arm actually did.	Closed the loop: read real wrist position from TF after every move and correct for the error.

31 Aug – 3 Sep — the hand's geometry had never been measured

Date	Bug	Fix
1 Sep	Grasp code aimed at the wrist position, not where the fingertips actually converge.	Built <code>finger_geometry_check.py</code> to measure the real fingertip offset via TF; corrected aiming to use it.
1–3 Sep	Impossible geometry: the hand's thumb-to-index span closes to only 9.11 cm; every test apple was 7.36–8.64 cm across — narrower than the hand's own closed span. No grip strategy could ever have worked.	Resized all ten apples to 5.00–6.00 cm radius; switched strategy from pinch to power/pocket grasp.
3 Sep	Several fingertip joints never closed — their <code>mimic</code> tags pointed at master joints that didn't exist in the URDF.	Fixed the mimic-tag references. Index fingertip travel went from 0.6 mm to 10.2 cm.
1–3 Sep	Apple reset via teleport kept up to ~2.4 m/s of leftover velocity — teleport moves position only, never velocity.	Added full delete-and-respawn on reset when re-teleporting won't settle it.
3 Sep	Couldn't tell "hitting the table" from "arm straining against gravity" — same symptom, different cause.	Built dedicated sweep scripts to separate the two.

6 Sep — the hand starts actually closing on the apple

Date	Bug	Fix
6 Sep	Open hand's fingertips hang 0.164 m below the wrist and ground out on the table before reaching grasp height.	Pre-curl the fingers to a partial pitch before descending, so they're retracted enough to clear the table.
6 Sep	False success — flinging: physics could launch an apple 79 m into the air; "height increased after lift" logged it as HELD.	Also require the apple to end up within one hand-width of the wrist, not just higher up.
6 Sep	False success — frame mismatch: apple position (world frame) compared directly against wrist position (robot frame); reported 0.447 m when the true distance was 0.108 m.	Convert both into the same coordinate frame before comparing.
6 Sep	False success — table miscounted as grasp: fingers jammed into the tabletop read 4/5 "contacts" at saturated ~100 Nm while the apple never moved.	Cross-check each fingertip's real TF position against the apple's known position/radius before counting a contact.
6 Sep	Closing at 0.05 rad/step produced 7–11 Nm peaks (~110 N) against a 0.12 Nm threshold — enough to launch a 180 g apple before most fingers arrived.	Slowed to small steps, checked 5×/step; contact threshold lowered to 0.075 Nm (idle noise floor is 0.043–0.046 Nm).
6 Sep	Thumb was never actively positioned — rested on the tabletop, reading 15.3 Nm vs 0.38–0.40 Nm on the other fingers.	Commanded a fixed thumb yaw. <i>Later found insufficient — see 7 Sep.</i>
6 Sep	Almost all apple displacement (up to 1.26 m) happened during descent, before any finger moved — the open hand was raking straight through the apple on the way down.	Approach from beside the apple: back off 0.16 m along the hand's forward axis, move in horizontally, then close.
6 Sep	Even with 0.000 m wrist error and a perfectly parallel palm, the hand still knocked the apple 0.14 m — the aim point itself sat inside the palm's own body.	Pushed the aim point 4.5 cm further along the hand's forward axis. <i>Resolved 7 Sep.</i>

7 Sep (afternoon) — where the apple sits in the hand, and the thumb fighting the table

Date	Bug	Fix
7 Sep	Every IK failure was reported as an orientation failure. One message read "orientation NOT achievable ... best axis match 1.000" — a <i>perfect</i> orientation match — when the real problem was that the position was out of reach.	Report which check actually failed (<code>0626d54</code>).
7 Sep	Fingers stopped the instant they felt the apple, so they touched it without ever holding it.	Added a squeeze phase that closes past first contact under a force cap (<code>79fc66f</code>).
7 Sep	The wrist correction fed back the full measured error and oscillated instead of converging — sequences like 0.062 → 0.034 → 0.036 that never settle.	Damped the correction to 60% of the measured error per iteration (<code>179bd97</code>).
7 Sep	False success — phantom squeeze: the squeeze phase reported a force reading for fingers that had never touched anything.	Squeeze only the fingers that actually registered contact (<code>179bd97</code>).
7 Sep	Finger joints were declared with a 100 Nm effort ceiling and could hit the apple hard enough to launch it.	Capped all 21 finger joints at 1.5 Nm in the URDF (<code>36ad5df</code>).
7 Sep	The palm-centre hypothesis, tested and disproved. Aiming the apple at the centre of the palm knocked it 0.697 m and 1.747 m before a finger moved, despite 6 mm wrist accuracy. See Table A.	Fixed the aim depth partway out along the fingers (<code>59e6eac</code>).
7 Sep	The thumb was stabbing the tabletop, not the apple. Every finger read its 1.50 Nm cap while the thumb read 18.4, 69.8 and 16.5 Nm. The thumb protruded 0.119 m below the palm, putting its tip 2 mm above the table. See Table B.	Swung the thumb to yaw −0.50 rad for 26 mm of clearance, keeping a 0.135 m thumb-to-finger gap (<code>c19a076</code>). Verified in the next run: thumb effort fell to 0.18–0.31 Nm.

7 Sep	One attempt ran out of correction iterations while still converging, and grasped 3.1 cm out of position.	Raised the correction budget from 5 to 9 (adc3997).
7 Sep	False success — a full run measured against a dead simulation. The TF tree had split, joint states had stopped, and the script closed the hand anyway.	Verify the robot state is readable before starting, and abort if it disappears mid-run (cc38b12).

7 Sep (evening) — tilting the palm, centring the apple, and the first complete pick

Date	Bug	Fix
7 Sep	The fingertip centroid (mean of five tips) cannot show whether the apple is centred; two attempts reported 5/5 contacts yet ended with only one finger loaded.	Measure each fingertip's own gap to the apple surface before closing (01c8362).
7 Sep	Closing the thumb <i>last</i> was too late: the four side-by-side fingers push the apple the same way, and with nothing opposing them it was shoved out before the thumb arrived.	Thumb closes first to contact as a backstop, then the fingers (931d109). A first-contact touch measured only 0.10–0.28 Nm, so a force-capped preload was added (3120154).
7 Sep	With the palm flat, the fingers could only reach the apple's top. They met it 41–55 mm above its equator (the pinky within 1 mm of the top pole; in one attempt ring and pinky were above the apple entirely), while the thumb met it 21 mm below — 65–76 mm apart, unable to oppose. A finger descending onto a sphere reaches only its top cap. See Table C.	Added a contact-height measurement (3120154) and made palm tilt the variable (e23c04c). At 30° the fingers met the apple at +3 to +5 mm with the thumb at –6 mm.
7 Sep	The thumb preload overshot its 1.2 Nm cap to 16.79 Nm: contact force jumps between two samples with nothing in between.	Finer preload steps, force re-read within each step (9bb83e5). <i>Still overshoots to 4.9–6.2 Nm; open.</i>
7 Sep	Hypothesis tested and disproved: that a moving finger's own effort exceeded the contact threshold, so fingers "contacted" in mid-air. Calibration measured 0.029 Nm moving in free air — no different from idle.	Kept the calibration, which showed the thumb is twice as noisy as the fingers (0.053 vs 0.029 Nm); contact thresholds became per-finger (0ea9a83 , 231f611).
7 Sep	Index and middle never touched the apple. They started 14–16 mm further from it than the pinky and peaked at 0.023 Nm, below free-air noise. The across-the-hand aim used the mean of all five fingertips, and the thumb's offset dragged the apple 14 mm toward the ring/pinky side.	Swept a lateral aim offset (2668fe4); +15 mm centred the apple to a 2 mm spread, and index and middle went from 0.023 to 1.500 Nm. See Table D.
7 Sep	Hypothesis tested and cleared: that the gripper camera's 4×4×2 cm collision box, mounted at the fingers' closing depth, blocked the pinky.	Measured fingertip clearance to the box: 63–116 mm. Not a factor (a8d7485).
7 Sep	The free-air calibration closes the hand onto itself: pinky and thumb read 1.500 Nm, the joint cap, with nothing in the hand, which set their contact thresholds to 3.000 Nm. A pinky that then reached 1.500 Nm on the apple was not counted.	Use the 90th percentile of samples below a self-collision level instead of the peak (8f3b465).
7 Sep	A parameter rename left four stale references; the pre-commit check compared names across the whole module, so a same-named parameter elsewhere hid them and a run died with <code>NameError</code> .	<code>tools/check_scopes.py</code> resolves names per scope with <code>symtable</code> ; verified by reintroducing the bug (4bb4bb0).
7 Sep	Hypothesis tested and found wrong, with the opposite result: that the squeeze pushed the apple out of the hand. Sweeping it showed the squeeze moved the apple only 1–10 mm, and more squeeze was strictly better. See Table E.	Squeeze 0.12 rad kept (9fab26). At 0.12 the apple was picked up and held: the first complete grasp (1533014).
7 Sep	Four identical repeats of the pick all failed. The arm settled the wrist 4–6 mm lower, which put index, middle and ring 3 mm <i>inside</i> the apple before closing instead of 3–6 mm outside; the apple then moved 0.110–0.211 m during closing against 0.029 m in the pick.	Tightened the wrist tolerance from 20 to 8 mm (6bb7771). <i>Later reversed — see 13 Sep.</i>
7 Sep	The settle check compared two position snapshots, and a bouncing apple is momentarily still at the top of each bounce: resets logging 0.4, 1.8 and 32 m of travel still reported "settled".	Require four consecutive still readings and the apple within 10 mm of its spot (4496744). Descent disturbance fell from 0.120–0.340 m to 0.002–0.006 m.
7 Sep	The correction loop stalled, reporting 0.014 m nine times: the arm at that reach was holding against gravity at its limit. The wrist also settled ~10 mm high, putting contacts back above the equator.	Detect a stalled correction and continue; aim the back-off lower (003948a).

13 Sep — the approach, the physics of the apple, and a regression found and reversed

Date	Bug	Fix
13 Sep	Across thirteen attempts, those making 0–1 correction moves beside the apple disturbed it 2–16 mm (including the pick), while those making 6–9 disturbed it 41–340 mm and all failed; the 8 mm tolerance had forced more such moves. Separately, fingertip gaps were measured against where the apple was placed, not where it was: one attempt reported the apple "centred" while it was 0.167 m away.	Corrections moved to a pre-grasp pose 40 mm back, final approach in straight-line steps; all checks switched to the apple's live position (2e2906a).

13 Sep	The next run was worse: the apple knocked 0.18–1.95 m. The cause was in <code>solve_ik</code> , which chose among valid solutions by total joint angle from zero and never considered the arm's current configuration, so nearby targets could return different arm configurations. Logged as correction errors of 0.023 → 0.078 → 0.077 → 0.023 m.	Seed IK from the current joints and choose the least joint motion; refuse moves needing a large joint swing; log every apple disturbance against the arm move that caused it (<code>9e77f3d</code>).
13 Sep	False readings in the output: a thumb-to-finger angle of "3°" computed about an apple 1.9 m away; "GRIP ... a working grasp configuration" printed for runs that lifted nothing; a closing message citing an 8.00 cm apple long after the apples were resized.	Opposition reported only once a finger is on the apple; contact reported as contact only; message corrected (<code>9e77f3d</code>).
13 Sep	With knocks now attributed, all fell in the straight-line approach. The first move lands 58–78 mm off, more than the 40 mm standoff; and a 0.35 rad swing limit refused every correction, since a 60 mm correction genuinely needs ~0.4 rad.	120 mm standoff, a 12 mm gate before approaching, and a 12 × 10 mm approach that stops if the apple moves at all (<code>b9a3ade</code>).
13 Sep	The apples rolled away on their own. Attempts stopped because the apple moved 11–18 mm while the hand was still 60–80 mm away; one moved 0.028 m across four consecutive arm moves — constant-speed rolling. The apple's collision shape was a perfect sphere, with a single contact point and nothing resisting a roll. The settle check also allowed up to 4 mm/s of drift.	A 50 mm-wide flat base on all ten apples, flush with the bottom so resting height and grasp contacts are unchanged; settle check judges the whole sampling window; disturbance log reports fingertip distance (<code>e8565e4</code>). Every reset in the following run settled first try. An XML comment containing "---" initially made all ten models unloadable, caught by parsing each file before commit.
13 Sep	<code>solve_ik</code> accepts answers up to 25 mm off (50 mm on fallback), each solve with a different error, so feeding the measured error back into another solve chased solver noise: corrections of 25, 37 and 13 mm were followed by 10 mm steps that missed by 20, 33 and 23 mm, and one attempt's accumulated target walked 117 mm to within 6 mm of the apple.	Jacobian-based servoing for small moves, capped at 10 mm and 0.06 rad per step, verified offline against the UR5e's published kinematics (Table F). Added results tables to the run output (<code>4402e6b</code>).
13 Sep	A regression, identified by comparing every run since the pick. The last three runs placed no finger on the apple at all: the approach limits had become stricter than the arm could meet (stopped at 16–19 mm against a 15 mm limit). See Table G.	Ran the approach that produced the pick side by side with the servo approach in the same run (<code>e2b6d67</code>).
13 Sep	The side-by-side test was decisive: the pick approach reached the apple 2/2, the servo 0/2 with its error unchanged at 125–127 mm for 25 steps. Separately, the grip reading used only the knuckle joint of each three-joint finger; the successful pick had logged four fingers at the idle floor while lifting the apple 8.5 cm, so earlier "fingers let go" readings were unfounded.	Servo dropped; pick approach kept with tolerance 12 mm and at most 3 corrections; grip reported as the strongest of each finger's three joints (<code>6b33941</code>).
13 Sep	Best run since the pick: apple reached 4/4, fingers in contact 5/5/5/3, lifts +5.9, +4.3, +2.0 and −0.2 cm. Sorted by lift, the height at which the fingers met the apple lines up exactly with the lift, and the two best attempts are the two where the arm happened to settle below its target. See Table H.	Aim the grasp 8 and 15 mm lower on purpose (<code>79d49c7</code>). <i>The run that followed disproved the relationship; see the next entry.</i>

13 Sep (afternoon) — from one pick to three in four

Date	Bug	Fix
13 Sep	Hypothesis tested and disproved: that contact height decides the lift. Aiming lower did exactly what it was meant to — all four attempts met the apple with 4 of 5 fingers below its equator, against 4, 1, 1 and 0 the run before — and the lift got <i>worse</i> : +3.7, +2.0, +1.8, +1.8 cm against +5.9, +4.3, +2.0, −0.2. Across seven attempts the fingers stayed loaded when the thumb pushed 5.5–6.2 Nm and lost their load in all four where it pushed 8.8–13.3 Nm, against a 1.2 Nm preload cap the code never enforced.	Grasp kept 8 mm lower. Enforced the thumb cap by backing the thumb off until its push is within the limit, tested against the uncapped thumb; "still gripping" now requires 0.30 Nm, above the 0.01–0.09 Nm idle floor (<code>c94f97c</code>).
13 Sep	The thumb-cap test was inconclusive (lifts +6.2 and +3.1 cm capped, +1.6 and +2.8 uncapped; the thumb still peaked at 12–23 Nm either way). More importantly, attempts with near-identical grasp geometry had now lifted the apple anywhere from +1.6 to +8.5 cm, so four-attempt comparisons of one change could not separate a real effect from chance, and nothing measured <i>before</i> the lift predicted its outcome.	Thumb cap kept as specified behaviour. Four identical attempts, with the lift itself instrumented: hand rise, apple rise, the gap between them, and finger loads, sampled throughout (<code>947fb67</code>).
13 Sep	Second complete pick (+8.5 cm), with the others at +6.4, +3.9, +3.9. The watch revealed something no earlier measurement could: after 6 s of a lift commanded to take 3 s, the hand had risen only 12–19 mm, and the apple had kept pace with it in every attempt. The apple was not slipping; the hand was not rising.	Watch the whole lift, up to 25 s, and record all six arm joint efforts against their declared limits (<code>0ebade3</code>).
13 Sep	Third complete pick , the highest yet at +11.3 cm. False reading — an arm failure scored as a grip failure: in two attempts the apple stayed within 5 mm of the hand throughout, but the hand rose only 2.3–2.4 cm with <code>wrist_1</code> at its 28 Nm limit, and both were scored NOT HELD. An attempt holding nothing lifted the hand 14.4 cm with <code>wrist_1</code> at 9 Nm. The slip threshold also reported the apple settling 6 mm into the fingers as a slip.	A separate ARM STALLED verdict; slip threshold 5 → 15 mm; empty-hand control attempts added (the apple is moved away just before the hand closes) to separate "holding the apple overloads the arm" from "the arm stalls at this pose anyway" (<code>ba1f815</code>).

13 Sep	Fourth complete pick , +14.7 cm. The controls lifted the empty hand 15.0 and 15.7 cm with wrist_1 at 7 Nm, while both grasps drove wrist_1 to 28 Nm. The apple's weight can add at most 2.6 Nm at that joint, so its weight alone could not explain the difference. See Table J.	Tested the largest remaining force in the grasp, the thumb squeeze (6.5–10 Nm at lift start): half the attempts eased the thumb to 3 Nm just before lifting (3f4551e).
13 Sep	Two picks in one run (+13.0 and +7.8 cm). Hypothesis tested and disproved : easing the thumb from 13.6 to 1.5 Nm did not help; the apple's weight pushed its load back to 6–8 Nm within 3 s and wrist_1 still reached 28 Nm. False reading — stalls that were not stalls : the hand rose in steps with pauses of about 2 s, and the watch ended after any 2 s pause or at 25 s, so it gave up on arms that were still lifting — one was cut off at 15 mm, one at 65 mm, and one was still rising at 5 mm/s when its 25 s ran out.	Thumb easing dropped. Time the lift from the simulation's own clock (the /joint_states stamp), watch up to 60 s, stop only after 8 s without rising, and log wrist_1 on every sample so a brief spike can be told from a joint held at its limit (214d6b5).
13 Sep	Best run of the project: three of four attempts picked, each lifted the full height (+16.7, +16.6, +16.7 cm), with no slip. The lift commanded for 3.0 s took 3.8 s of simulated time but 45–48 s on the wall clock: Gazebo runs at about 8% of real time on this machine , which is why the lift looked slow and why the wall-clock watch misjudged it. wrist_1 touched 28 Nm only as a single-sample spike — at its limit in 0% of samples. See Table K.	The arm-stall problem is closed: it was a measurement artefact. The remaining failure (attempt 1: first move 96 mm off, corrections disturbed the apple 43 mm, no grip) was also the approach failure in the previous run's first attempt, and is the next target.
13 Sep	The identical repeat also picked 3 of 4 (+16.7, +16.6, +16.8 cm), meeting the apple_06 acceptance criterion of at least 6 of 8. In the last three runs the only failures were attempt 1: its first move measured 84, 96 and 94 mm off, against 44–75 mm for attempts 2–4. Wall-clock waits after arm moves (20 s for a 3.0 s move) were only about 1.7 s of simulated time, and the reset between attempts waited about 0.4 s simulated. So the arm was measured mid-move, and attempts 2–4 set off from beside the apple while attempt 1 started from a true rest pose.	Wait out every arm move, and the reset, in simulated time (7e769f3). <i>Reversed the next day; see 14 Sep.</i>

14 Sep — timing, frozen simulations, and all ten apples

Date	Bug	Fix
14 Sep	A fix that broke the grasp. With every move waited out, the arm landed exactly on target (first move 0 mm), which confirmed the earlier 44–107 mm "misses" were mid-move measurements. But picks fell to 0 of 4 . Once the arm really reached the 0.16 m back-off pose, its joint-space swing to the grasp pose drove a finger through the apple (72 mm disturbance in two attempts, 28 mm in a third), and each attempt took 250–630 s instead of 130–170 s. The working approach had depended on the arm cutting that corner. In one attempt Gazebo froze outright: 0.0 s of simulated time over 8 s of lift, scored ARM STALLED.	Reverted (e19a795). A lift during which the simulation clock does not advance is now scored SIM FROZE.
14 Sep	False reading — a slow simulation scored as an arm stall. The reverted, previously 6-of-8 code picked 0 of 4 on a fresh run. Yet two attempts held the apple with all four fingers loaded and 0–1 mm lag. While lifting those tight grips (thumb 10–13 Nm, fingers at their 1.5 Nm cap) the simulation ran at 0.017× and 0.020× real time, against 0.089× with an empty hand in the same run. So the 60 s watch ended when the arm had had only 1.0 s and 0.6 s of its 3.0 s move.	Watch the lift until the arm has had 3.3 s of simulated time, however long that takes on the wall clock, and report the real-time factor. A watch that runs out first is scored SIM TOO SLOW (b3b1b72).
14 Sep	Next run, unchanged grasp: 3 of 4 (+16.3, +16.5, +16.4 cm) at a normal 0.083–0.091×. Across three runs attempts 2–4 had now picked 9 of 9 and attempt 1 none. Attempt 1 failed exactly as before: first move measured 104 mm off, and three corrections each disturbed the apple 14–19 mm with fingertips 6–11 mm away.	Wait out <i>only</i> the grasp move in simulated time, leaving the approach move untouched. A/B tested in one run: waiting on attempts 1 and 3, unchanged on 2 and 4 (91aaa7a).
14 Sep	The A/B run crashed before the first grasp move with UnboundLocalError. A new helper, sim_now(), shared its name with a variable assigned later in the same function, which makes the name local throughout that function. The scope checker only looked for undefined names, so it passed.	Helper renamed; tools/check_scopes.py now also flags a local variable that hides a module-level function, and was verified to catch exactly this case (be2ce42).
14 Sep	First 4-of-4 run. With the grasp move waited out (attempts 1 and 3): 0 mm off, no corrections, apple disturbed 8 mm, wrist_1 peak 12 Nm. Without it (2 and 4): 40–50 mm off mid-move, one correction disturbing the apple 10–13 mm, and wrist_1 at 28 Nm and shoulder_lift at 150 Nm at the very start of the lift. This explained the joint-limit readings seen since 13 Sep: they were the arm still completing an interrupted correction, not the load of the grasp. Attempt 1, which had failed in every earlier run, picked. See Table N.	Waiting out the grasp move made the default (4481eb4). Confirmation run: 4 of 4 again , every attempt 0 mm off, apple rose +15.1–15.2 cm (spread 0.1 cm), no joint at its limit. apple_06: 8 of 8 over the two runs.
14 Sep	Size extremes, settings unchanged. apple_01 (10.0 cm, 0.41 kg): 4 of 4 , +15.1–15.2 cm, apple disturbed 1 mm. apple_10 (12.0 cm, 0.70 kg): 4 of 4 , +14.2–14.4 cm. It had thinner margins: a fingertip passed 5 mm from the larger apple on the way in, disturbing it 17 mm, and the heavier apple sat 6–8 mm lower in the hand during the lift. Because the hand aims at each apple's live centre, its position follows the apple's size without scaling.	Optional attempt count added to the test, and test_all_apples.sh written to run the remaining seven apples in sequence (40da017).

14 Sep	All ten apples picked. Apples 02, 04, 05, 07, 08 and 09 each picked 2 of 2; apple_03 1 of 2 (Table M). False reading — a frozen simulation scored as a failed grasp: in apple_03's second attempt the simulation clock advanced 0.0 s during a 300 s wait. The arm never moved (164 mm off before and after three corrections), and carrying on drove the hand into a meaningless pose. The runner's summary also flagged every apple as having a simulation problem, because it matched the verdict names printed in each run's legend.	An attempt whose grasp move gets less than 3.0 s of simulated time is abandoned and scored SIM FROZE (10c32ed). The runner now matches only real verdicts, verified against a legend-only log and a real one (c7399dc).
--------	--	---

6.1 Measurements behind the September entries

Table A — where in the hand should the apple sit? (7 Sep, apple_06)

Aim depth into the hand	Fingers in contact	Apple moved during descent	Wrist error
0.060 m — against the palm	0 / 5	0.697 m	0.006 m
0.085 m	0 / 5	1.747 m	0.028 m
0.110 m — adopted	3 / 5	0.008 m	0.007 m
0.128 m — at the fingertips	0 / 5	0.000 m	0.013 m

Table B — the thumb against the tabletop (7 Sep)

Thumb yaw	Protrudes below palm	Tip height	Table clearance	Thumb-to-finger gap
0.00 rad — as run	0.1190 m	0.402 m	+2 mm	0.179 m
−0.50 rad — adopted	0.0946 m	0.426 m	+26 mm	0.135 m

Table C — palm tilt and where the fingers meet the apple (7 Sep; + above the equator, − below)

Palm	Index	Middle	Ring	Pinky	Thumb	Fingers vs thumb
Flat	+41 mm	+44 mm	+52 mm	+55 mm	−21 mm	~70 mm apart
Tilted 30°	+5 mm	+3 mm	+4 mm	+4 mm	−6 mm	~10 mm apart

The apple's radius is 55.5 mm, so +55 mm is its top pole. 45° was later adopted: it is the tilt at which fingers first met the apple below its equator once back-off and lateral centring were set.

Table D — centring the apple across the four fingers (7 Sep, tilt 45°)

Lateral aim offset	Spread of fingertip gaps	Index / middle peak effort	Apple lift
0 mm (mean of five tips)	11 mm	0.023 Nm	+0.000 m
−15 mm	24 mm	0.023 Nm	+0.000 m
+15 mm — adopted	2 mm	1.500 Nm	+0.043 m
+30 mm	11 mm	1.500 Nm	+0.048 m

Table E — the squeeze sweep that produced the first pick (7 Sep, pre-shape 0.4)

Squeeze past first contact	Fingers in contact	Contacts below equator	Apple moved by the squeeze	Apple lift
0.00 rad	0 / 5	—	0.001 m	+0.000 m
0.02 rad	1 / 5	1 / 1	0.010 m	+0.000 m
0.05 rad	3 / 5	0 / 3	0.010 m	+0.000 m
0.12 rad	5 / 5	4 / 5	0.007 m	+0.085 m — HELD

Table F — offline verification of the Jacobian servo (13 Sep)

Test, against published UR5e kinematics with a simulated 30 mm sag and 90% command tracking	Result
Correct a 75 mm position error	Reached 1.9 mm in 8 steps; largest joint step 0.017 rad
Follow 12 × 10 mm approach waypoints	All reached; at most 2 steps per waypoint
Correct a palm tilted 10°	Corrected to 0.61°. The test initially failed, exposing that the stop condition ignored orientation; fixed
Stop when the apple moves	Aborted after the first step

The servo passed offline but made no progress on the simulated arm (Table G), which the offline model does not capture. It was therefore dropped on the evidence of the live runs.

Table G — fingers touching the apple, per attempt, in every run since the pick

Commit	Change under test	Fingers touching (4 attempts)	Best lift
9fab26	Squeeze sweep	0, 1, 3, 5	+8.5 cm HELD
1533014	Repeat of the pick	2, 2, 1, 0	–
4496744	Bouncing-apple check	0, 5, 3, 2	+1.2 cm
003948a	Aim lower, stall detection	4, 0, 0, 0	–
9e77f3d	One arm configuration	4, –, –, –	–
b9a3ade	Approach gate	–, –, –, –	–
e8565e4	Flat apple base	–, –, –, –	–
4402e6b	Jacobian servo	–, –, –, –	–
e2b6d67	Pick approach vs servo, alternating	5 (pick), – (servo), 2 (pick), – (servo)	+1.4 cm
6b33941	Pick approach, 12 mm tolerance, 3 corrections	5, 5, 5, 3	+5.9 cm

– means the hand never reached the apple, so no finger could touch it.

Table H — how low the fingers wrap decides the lift (13 Sep run, plus the 7 Sep pick)

Attempt	Wrist height (target 0.604 m)	Where the four fingers met the apple	Average	Below equator	Apple lift
7 Sep pick	0.597 m	–16, –19, –9, –5 mm	–12 mm	4 / 5	+8.5 cm
13 Sep, #1	0.587 m	–10, –14, –11, –7 mm	–10 mm	4 / 5	+5.9 cm
13 Sep, #2	0.601 m	+2, –6, +1, +3 mm	0 mm	1 / 5	+4.3 cm
13 Sep, #3	0.601 m	+3, –7, +2, +4 mm	0 mm	1 / 5	+2.0 cm
13 Sep, #4	0.609 m	+11, +7 mm (2 fingers)	+9 mm	0 / 3	–0.2 cm

Contacts below the apple's widest point carry it upward; contacts above it press it into the table. The two best attempts were the two where the arm settled below its target, which motivated aiming the grasp lower deliberately. **The next run disproved this as the deciding factor** (Table I, first row): with 4 of 5 contacts below the equator in every attempt, lifts fell. Contacts below the equator remain necessary — all nine picks have them — but they are not sufficient.

Table I — every run on 13 September after the aim-lower change (apple_06, 4 attempts each)

Commit	Change under test	Apple rose, per attempt	Picked
79d49c7	Grasp aimed 8 / 15 mm lower	+3.7, +1.8, +2.0, +1.8 cm	0 / 4
c94f97c	Thumb preload cap enforced vs. not	+6.2, +1.6, +3.1, +2.8 cm	0 / 4
947fb67	Identical attempts; lift watched for 6 s	+8.5, +6.4, +3.9, +3.9 cm	1 / 4
0ebade3	Whole lift watched (25 s), arm joint efforts	+0.0, +11.3, +3.7, +3.8 cm	1 / 4
ba1f815	Grasps alternated with empty-hand controls	+14.7, +2.8 cm (2 controls)	1 / 2
3f4551e	Thumb eased before lift vs. not	+0.0, +2.4, +7.8, +13.0 cm	2 / 4
214d6b5	Identical attempts; lift timed in simulated time, watched to the end	–40.0, +16.7, +16.6, +16.7 cm	3 / 4

Picked means the apple rose at least 7.5 cm and ended within 0.20 m of the wrist. In 0ebade3–3f4551e several attempts kept the apple in the hand but were cut short by the lift watch (Table K); the –40.0 cm in 214d6b5 is an apple knocked off the table during the approach.

Table J — empty-hand control: what loads the wrist? (ba1f815)

Attempt	Hand rose	wrist_1 peak (limit 28 Nm)	shoulder_lift peak (limit 150 Nm)	Apple rose
Grasp	1.8 cm*	28 Nm	150 Nm	+2.8 cm
Empty-hand control	15.0 cm	7 Nm	–	–
Grasp	12.7 cm*	28 Nm	101 Nm	+14.7 cm
Empty-hand control	15.7 cm	7 Nm	–	–

* Under the 25 s wall-clock watch later shown to cut lifts short. The peak-effort difference is real, but Table K shows it is a momentary spike, not a joint held at its limit. Table N (14 Sep) shows where the spike comes from: it appears only when the lift starts while the arm is still finishing a correction move.

Table K — the best run, with the lift timed in simulated time (214d6b5)

Attempt	First move off	After correction	Apple disturbed	Fingers touching	Hand rose	Lift time: simulated / wall	wrist_1 at limit	Apple rose
1	96 mm	30 mm	43 mm	4 / 5, none gripping	17.2 cm	3.8 s / 37.5 s	0%	knocked off
2	75 mm	7 mm	5 mm	5 / 5	16.9 cm	3.8 s / 48.1 s	0%	+16.7 cm

3	70 mm	7 mm	12 mm	5 / 5	17.0 cm	3.8 s / 44.8 s	0%	+16.6 cm
4	61 mm	7 mm	10 mm	5 / 5	16.9 cm	3.8 s / 47.5 s	0%	+16.7 cm

The commanded lift is 3.0 s and 15 cm; the hand overshoots slightly to about 17 cm. A simulated-to-wall ratio of 3.8 s to 45–48 s is a real-time factor of about 0.08. The overshoot to 17 cm and the wrist_1 limit readings both disappear once the grasp move is waited out (Table N).

Table L — every run on 14 September (4 attempts unless stated)

Commit	Apple	Change under test	Apple rose, per attempt	Picked
214d6b5	apple_06	Identical repeat (13 Sep, late)	+0.0, +16.7, +16.6, +16.8 cm	3 / 4
7e769f3	apple_06	Every arm move waited out in simulated time	+0.1, +0.0, +0.0, +0.0 cm	0 / 4
e19a795	apple_06	Reverted; Gazebo running slowly and freezing	+0.0, +5.5*, +3.0*, +0.2* cm	0 / 4
b3b1b72	apple_06	Lift watched in simulated time	+0.0, +16.3, +16.5, +16.4 cm	3 / 4
be2ce42	apple_06	A/B: grasp move waited out on 1 and 3	+15.2, +16.6, +15.1, +16.7 cm	4 / 4
4481eb4	apple_06	Waited out on every attempt	+15.2, +15.1, +15.2, +15.2 cm	4 / 4
4481eb4	apple_01	Smallest apple, settings unchanged	+15.2, +15.2, +15.1, +15.2 cm	4 / 4
40da017	apple_10	Largest and heaviest apple, settings unchanged	+14.2, +14.3, +14.3, +14.4 cm	4 / 4
40da017	02–05, 07–09	Remaining seven apples, 2 attempts each	see Table M	13 / 14

* The apple was still held when the watch ended; the simulation had given the arm only 0.1–1.0 s of its 3.0 s lift.

Table M — all ten apples, one set of grasp settings (13–14 Sep)

Apple	Diameter	Mass	Friction μ	Picked	Apple rose (hand 15.0 cm)	Apple moved, whole attempt
apple_01	10.0 cm	0.41 kg	0.80	4 / 4	+15.1 to +15.2 cm	1 mm
apple_02	10.2 cm	0.40 kg	0.85	2 / 2	+15.1 to +15.2 cm	2–3 mm
apple_03	10.4 cm	0.48 kg	0.75	1 / 2 †	+15.2 cm	4 mm
apple_04	10.7 cm	0.47 kg	0.90	2 / 2	+14.9 to +15.3 cm	3–10 mm
apple_05	10.9 cm	0.47 kg	0.70	2 / 2	+15.2 to +15.3 cm	11–14 mm
apple_06	11.1 cm	0.54 kg	0.85	8 / 8	+15.1 to +15.2 cm ‡	8–9 mm
apple_07	11.3 cm	0.55 kg	0.95	2 / 2	+15.3 cm	9–14 mm
apple_08	11.5 cm	0.65 kg	0.78	2 / 2	+14.4 to +14.8 cm	15–17 mm
apple_09	11.8 cm	0.59 kg	0.90	2 / 2	+14.8 to +15.1 cm	17–18 mm
apple_10	12.0 cm	0.70 kg	0.82	4 / 4	+14.2 to +14.4 cm	17 mm
All	10.0–12.0 cm	0.40–0.70 kg	0.70–0.95	29 / 30	+14.2 to +15.3 cm	1–18 mm

† The miss was a frozen simulation (0.0 s of simulated time during the grasp move); the grasp was never attempted. ‡ Over the two runs with the grasp move waited out. Two trends follow size and mass. Larger apples are disturbed more on the way in, because the fingertips pass closer to them. Heavier apples sit a few millimetres lower in the hand during the lift. Neither caused a slip.

Table N — waiting out the grasp move, A/B in one run (be2ce42, apple_06)

Attempt	Grasp move waited out	First move off	Corrections	Apple disturbed	wrist_1 / shoulder_lift peak	Hand rose	Apple rose
1	yes	0 mm	0	8 mm	12 / 96 Nm	15.0 cm	+15.2 cm
2	no	50 mm	1	13 mm	28 / 150 Nm	16.7 cm	+16.6 cm
3	yes	0 mm	0	8 mm	12 / 99 Nm	15.0 cm	+15.1 cm
4	no	40 mm	1	10 mm	28 / 150 Nm	16.8 cm	+16.7 cm

Limits are 28 Nm (wrist_1) and 150 Nm (shoulder_lift). Without the wait, both joints reach their limits in the first sample of the lift, while the arm is still completing the correction; the extra hand rise is that correction's overshoot.

7. Best Result Achieved To Date

All ten apples picked with one set of settings — 13–14 September, commits 4481eb4–c7399dc

Palm tilted 45°, back-off 0.090 m, lateral aim +15 mm, grasp aimed 8 mm low, pre-shape 0.4, squeeze 0.12 rad, thumb yaw –0.50 rad, thumb preload capped at 1.2 Nm, grasp move waited out in simulated time. The same values for every apple. Per-apple results are in Table M; the figures below are for `apple_06`, attempt 1 of commit 4481eb4, which the other three attempts matched to within 1 mm of apple rise.

Measure	Achieved	Assessment
Apples picked, all ten	29 / 30 attempts	The one miss was a frozen simulation
Range covered	10.0–12.0 cm, 0.40–0.70 kg, μ 0.70–0.95	No per-apple tuning
Wrist position error when the grasp move finished	0.000 m	No correction needed
Fingertip gaps to the apple before closing	+10, +10, +8, +15 mm	All outside the apple
Apple disturbed during approach	0.008 m	Negligible
Fingers in contact	5 / 5	Complete
Hand rose / apple rose	15.0 cm / +15.2 cm	The full lift; apple stayed within 3 mm of the hand
Arm joint peaks during the lift	shoulder_lift 100 Nm, wrist_1 16 Nm	Below limits (150 / 28 Nm)
Spread of apple rise across four identical attempts	0.1 cm	Repeatable
Verdict	HELD	Every attempt, every apple

Every false-success guard in the pipeline had to pass for these verdicts. The apple rose more than half the commanded lift and stayed within a hand-width of the wrist, with both compared in the same coordinate frame. Contacts were confirmed by fingertip position rather than effort alone. The apple was settled before the attempt. The simulation was publishing joint states and its clock was advancing throughout. The lift was watched until the arm had had its whole move in simulated time. The first complete grasp, on 7 September (commit 9fab26), lifted `apple_06` 8.5 cm under the same hand settings except the 8 mm lower aim, the thumb cap and the waited-out grasp move.

Best figure recorded for each individual measure

These come from different attempts; no single attempt holds all of them at once.

Measure	Best recorded	Context
Complete grasps (picked and held)	44	1 on 7 Sep; 11 on 13 Sep; 32 on 14 Sep (Tables I and L).
Apples picked	10 / 10	14 Sep, one set of settings (Table M).
Picks in one run	4 / 4	14 Sep, four times: <code>apple_06</code> twice, <code>apple_01</code> , <code>apple_10</code> .
Consecutive picks, no misses	19	14 Sep, from the first 4-of-4 run through <code>apple_03</code> 's first attempt; the next attempt was the frozen simulation, and the ten attempts after it all picked.
Highest lift	+16.8 cm	13 Sep; the hand overshot to 16.8 cm before the grasp move was waited out.
Apple kept in the hand once grasped	every attempt	13–14 Sep: no apple whose fingers gripped it has slipped out during a lift.
Wrist position error	0.000 m	First on 6 Sep, where accuracy alone was shown not to be sufficient; every attempt since the grasp move was waited out on 14 Sep.
Palm alignment	1°	Routinely within 1–3° of the commanded tilt.
Apple disturbance, whole attempt	0.000 m	Achieved several times.
Apple resets settling first try	4 / 4	13 Sep, after the flat base was added; previously 2–3 teleports and a respawn were typical.
Fingertip travel (index)	10.2 cm	Was 0.6 mm before the dead <code>mimic</code> tags were fixed.

Assessment. The grasp works on every apple in the set with one configuration found by measurement. It is repeatable to within a few millimetres of lift, and no gripped apple has slipped. The last obstacles were not in the hand at all but in timing. Against a simulation running at 1.5–9% of real time, wall-clock waits measured the arm mid-move, provoked corrections that knocked the apple, cut lifts short, and made a frozen simulation look like a failed grasp. Each of these was found by logging the simulation's own clock alongside the motion. The work that remains is integration: carrying these settings into the vision-driven main pipeline and the place-in-crate step.

8. Current Status

Working and verified by measurement

- **All ten apples picked with one set of settings: 29 of 30 attempts**, lifted +14.2 to +15.3 cm, no slips. The one miss was a frozen simulation (Table M).
- `apple_06`: 8 of 8 over two identical runs, every attempt within 0.1 cm of the others.
- The arm lands on the grasp pose to within 1 mm once its move has finished; no corrections are needed, and the apple is disturbed 1–18 mm on the way in.
- Palm tilt (45°), lateral centring (+15 mm), squeeze (0.12 rad) and grasp height (8 mm low) are each settled by measured runs. The aim follows each apple's live centre, so it adapts to size without scaling.
- Every wait on the arm uses the simulation's own clock. Lifts are judged only after the arm has had its whole move, and a stalled or frozen simulation is reported as such rather than as a grasp or arm failure.
- Arm joints stay below their limits during the lift (shoulder_lift 90–103 Nm of 150, wrist_1 8–26 Nm of 28).
- Apple resets are reliable: the flat base stops constant-speed rolling, and the settle check rejects bouncing apples.
- Contact thresholds are calibrated per finger at the start of every run, robust to the hand touching itself.
- IK stays in one arm configuration between nearby targets; every apple disturbance is attributed to the arm move that caused it.
- Each run ends with results tables and a per-attempt verdict; `test_all_apples.sh` runs any set of apples and summarises them.
- Thirteen distinct false-success and false-reading defects are guarded against (Section 9.4); a pre-commit scope check covers two classes of runtime error that reached live runs.
- VLM fragility scoring runs end-to-end into a real force calculation.

Open

- **Integration.** The working settings are in `pocket_grasp_test.py` and have not yet been carried into the main pipeline, `full_layer_grasp.py`. So the VLM fragility layer and the place-in-crate step have not yet been run together with the working grasp.
- **Simulation speed and freezes.** Gazebo runs at 1.5–9% of real time on the development machine and occasionally stops outright, so an attempt takes 2–4 minutes and some runs need a restart. Results are unaffected but runs are slow.
- Only 1–4 of the 5 contacts are below the apple's equator, and the thumb meets the apple 18–20 mm above it, 95–140° from the fingers rather than directly opposite. The grasp holds, but it is not a geometrically opposed grip.
- Margins are thinnest on the largest, heaviest apples. A fingertip passes 5 mm from the apple on the way in (17–18 mm disturbance), and the apple sits 6–8 mm low in the hand. Apples outside the tested 10–12 cm range have not been tried.
- The smaller apples ran 2 attempts each; more repetitions would tighten the reliability estimate per apple.
- The learned closing policy was not needed for these results and has only a handful of verification generations. Nothing has been tested outside simulation.

9. Novelty

I searched the current published literature — arXiv, the DexHand project's own pages, and the ikpy library's documentation — specifically to check whether this project's central idea already existed before claiming anything. It largely doesn't, in the exact combination used here. Below is the headline finding, followed by four supporting technical contributions, each with the source checked.

9.1 Headline claim

PRINCIPAL NOVELTY CLAIM

This is the first documented case of a local, open-weight vision-language model driving fragility-conditioned grip force on a multi-fingered dexterous hand, and the first documented use of the DexHand v2 platform for any form of autonomous, perception-driven grasping.

The claim rests on two facts, each checked independently against the published record.

Fact 1 — no published research uses the DexHand v2 for autonomous grasping. The DexHand platform (dexhand.org, maintained by iotdesignshop / TheRobotStudio) has exactly one academic paper built on it that I could find: THETA (2026), which uses the earlier DexHand V1 purely for teleoperated hand-state estimation — a human drives it, the paper studies tracking accuracy, and it never touches autonomous perception or grasp planning. No published work uses DexHand v2 specifically, and none uses any version of DexHand for perception-driven, autonomous grasping of any object.

Fact 2 — no published VLM-conditioned grasp-force system runs on a fully local, small model or on a multi-fingered hand. The closest published system, DeliGrasp (2024), infers object properties for grasp-force selection using GPT-4 / GPT-4V — a large, closed-source, cloud-hosted model reached over an API — and applies it to a two-finger parallel gripper with one squeeze axis. This project instead runs a compact, open-weight 3B-parameter model (Qwen2.5-VL) entirely locally via Ollama, with no internet dependency or per-call cost, and applies the result to a 21-joint, five-fingered hand executing a multi-contact pocket grasp — a materially different control problem from squeezing two plates together. Recent 2026 work on edge-deployed VLMs for robots (e.g. arXiv:2601.14921) confirms local/edge VLM deployment is an active concern for robotics generally — but I found no instance of it applied specifically to grasp-force/fragility estimation, and none paired with a dexterous multi-fingered hand.

Taken together, the two facts leave the combination used here — a local open-weight model, conditioning grip force, on a five-fingered hand — unoccupied in the searchable published record. It is a genuine, citable gap rather than a restatement of an existing result. The qualifier that belongs on it: this is a claim about the *published record as searched*, and the system itself is demonstrated in simulation only, with complete grasps on all ten apples of the test set (29 of 30 attempts) using a grasp tuned by measurement; the vision layer has not yet been run together with that grasp.

9.2 Supporting contribution 1 — inverse kinematics for branching, multi-fingered end effectors

CLAIM

Solving IK for an arm terminating in a branching multi-fingered hand requires pruning every link downstream of the hand's root before the chain is built — a failure mode the ikpy library does not document and no published source appears to address.

ikpy's own wiki admits the underlying gap without solving it: "If your robot is not a linear chain (for example a humanoid), you must extract your chain in your URDF file" via `base_elements`, with no explanation of the mechanism. An independent user guide separately confirms "a common problem with IKPY is that it doesn't consider end effectors" properly. Neither source addresses what happens when the declared end-effector link still has branches hanging off it — exactly the DexHand case, where ikpy silently solved for the thumb tip instead of the wrist (Section 6, 28 Aug). The fix — programmatically pruning every joint/link downstream of the hand's root before building the chain, so the wrist is left as the one unambiguous terminal link — is not documented or published anywhere I could find, for this or any other branching-hand-on-arm combination.

9.3 Supporting contribution 2 — grip geometry derived from live measurement

CLAIM

An off-the-shelf hand's usable grasp geometry cannot be read from its description package and must be measured from the running system — a reusable method, and one that in this project repeatedly overturned conclusions that reasoning alone had got wrong.

The DexHand v2's own description package states joint limits, not where the fingertips actually converge at full closure. That pocket centroid, and the discovery that the hand's true closed span (9.11 cm) was narrower than every original test apple (7.36–8.64 cm) — meaning no grasp strategy could ever have worked before the apples were resized — came only from instrumenting the live simulation via TF. The same method produced the first complete grasp: palm tilt, lateral centring and squeeze were each settled by sweeping real attempts, and on four occasions the measurement contradicted the prevailing hypothesis — the palm centre, the squeeze pushing the apple out, finger motion registering as contact, and the camera body blocking the pinky. The method corrects itself as well as the design: on 13 September measurement disproved two further conclusions that had looked well supported — that contact height below the equator decides the lift (Table I), and that the thumb's squeeze was overloading the arm — and an empty-hand control showed that the apparent arm overload was specific to holding the apple before simulated-time logging showed it to be a momentary spike (Tables J and K). On 14 September the same practice located the last failures in timing rather than in the hand. Logging the simulation's own clock next to the arm's motion showed that the arm was being measured and corrected while still moving, and that the joint-limit readings were an interrupted correction (Table N). It also showed that a well-supported fix, waiting out every move,

broke the grasp by changing the approach path. The result was one configuration that picks every apple in the set (Table M). This measured-not-assumed approach is a reusable method for adapting any hobbyist or open-source hand description to a task it wasn't originally demonstrated on.

9.4 Supporting contribution 3 — silent false success and false readings, found and fixed

CLAIM

Thirteen distinct instances of silent false success or false measurement — a failure class the literature has only recently begun to formalize — were found and fixed in a live manipulation pipeline, each identified by direct measurement rather than by a benchmark.

"Silent" false-success in simulated manipulation is an active, named research concern — a 2026 study (arXiv:2606.03134) specifically measures how often it happens and how hard it is to detect. This project independently found and fixed thirteen concrete instances of that class (Section 6):

1. an apple physically flung 79 m logged as a successful hold;
2. a world-frame/robot-frame mismatch that under-reported distance by a factor of four;
3. table contact counted as a finger grasping the apple;
4. a squeeze phase reporting force for fingers that had never touched anything;
5. a full run of grasp measurements taken against a simulation that had stopped publishing joint states;
6. a settle check that reported a bouncing apple travelling metres between readings as "settled";
7. fingertip gaps measured against the apple's placed rather than live position, reporting it "centred" while it was 0.167 m away;
8. a grip reading taken from one of each finger's three joints, which reported the fingers of the one successful grasp as holding nothing;
9. a run summary announcing "a working grasp configuration" on runs that lifted nothing;
10. apples that stayed in the hand throughout the lift scored as dropped, because the hand had not risen far enough, blaming the grip for what appeared to be an arm failure;
11. "arm stalls" that were the test giving up: the lift was timed on the wall clock while the simulation ran at about 8% of real time, so arms that were still lifting were declared stopped;
12. an arm measured while still moving, reported as a 40–107 mm positioning error that the correction loop then "fixed" by pushing the hand into the apple;
13. a frozen simulation, whose clock had stopped, scored as an arm stall and as a failed grasp.

Fixing these by direct measurement, in a live pipeline rather than a benchmark study, is a concrete applied contribution to a problem the research community is only now starting to formalize.

9.5 Supporting contribution 4 — an outcome-scored, real-attempt closing policy search

CLAIM

The closing policy is searched against real physical outcomes — rewarding the apple's actual lift height rather than a proxy for contact — using a search strategy chosen to match a setting where each sample costs 60–120 seconds of live simulation.

The reward function driving the per-finger closing-policy search doesn't just reward contact — it weights genuine multi-finger contact, gives a large bonus specifically for the apple's height increasing during the lift (the strongest available signal of a real grasp), and penalizes both dangerous effort and uneven loading across fingers. Using a $(1+\lambda)$ evolution strategy — chosen deliberately because each sample is an expensive 60–120 second real Gazebo attempt, not a cheap simulated rollout — is a properly-matched, small-scale application of a known search technique to a genuinely hard-to-sample problem. It was built as the route to per-apple tuning. It turned out not to be needed for the current apple set, where one measured configuration picks all ten, so it remains available for objects outside that range.

9.6 Sources checked

- DexHand project (platform novelty is based against this): dexhand.org, github.com/iotdesignshop/dexhandv2_description
- THETA — the only paper found using any DexHand version, for teleoperation (not autonomous grasping): arxiv.org/pdf/2601.07768
- DeliGrasp — GPT-4/GPT-4V-based object-property inference for grasp policies, 2-finger gripper: arxiv.org/pdf/2403.07832
- Exp-Force — experience-conditioned pre-grasp force selection with VLMs: arxiv.org/pdf/2603.08668
- Edge-deployed VLMs for robotic perception, 2026: arxiv.org/pdf/2601.14921
- ikpy wiki — URDF import and the `base_elements` limitation: github.com/Phylliade/ikpy/wiki/URDF
- Independent IKPy/Webots guide noting the end-effector handling gap: gist.github.com/ItsMichal
- Silent manipulation failures / false-success detectability study: arxiv.org/html/2606.03134

Compiled directly from the project's source code and commit history (c7399dc back to the initial commit), 14 September 2026. Measurements quoted are from live Gazebo runs logged by `pocket_grasp_test.py`, `thumb_opposition_check.py` and `finger_geometry_check.py`, not from specification values. The Jacobian servo verification (Table F) is offline, against published UR5e kinematics.